ACADEMIC RESEARCH DOCUMENTATION

BUILDING STOCK ENERGY WORKBENCH

Academic User Guide

Author: Mirza Saribiyik

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Building Stock Energy Workbench (BSEW) is a local research interface for preparing, executing, auditing and exporting reproducible building-stock energy simulations. This guide follows the operational path **Files → preflight → Run → safe stop/resume → Outputs → building and spatial evidence → quality assurance → provenance and citation**.

> **Core interpretation rule.** BSEW produces modelled site-energy and operational-emissions evidence under recorded assumptions. It does not report measured utility consumption, building-specific calibration, certification, or a prediction guaranteed to match future operation.

Suggested citation

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1. Purpose, audience and reading path

BSEW is intended for a mixed academic audience: researchers who need a clear main workflow, and technical reviewers who need access to model, simulation, diagnostic and provenance evidence. The main interface therefore has three operational surfaces: Files, Run and Outputs. Expert files remain available as evidence but are not the primary interpretation path.

Use this guide in three ways:

- Sections 2-9 support a first defensible run.
- Sections 10-18 explain outputs, QA and building-level evidence.
- Sections 19-26 establish QA, provenance, limitations, reporting and citation.

The guide does not replace a study protocol. Before running, define the research question, spatial scope, scenario, expected period, exclusion policy, retention level and acceptance criteria. A run name is not a protocol; it is one identifier within a larger evidence chain.

The main worked example in the final edition will be the completed annual run `ALL_VALENC-A_REAL`. No interim total from the running ledger is published here. Screenshots will retain the real run and building identifiers while masking local paths and user information.

1.1 Terminology

Run means one named stock execution with frozen inputs and profile identity. **Building record** means the latest ledger record for a cadastral reference within that run. **Aggregate** means the derived stock summary. **Evidence** means the preserved inputs, identities, model artefacts, EnergyPlus outputs, QA decisions and exports needed to audit a claim.

2. Scientific scope and non-claims

BSEW transforms cadastral and research inputs into simplified OpenStudio/EnergyPlus building models, executes them under a verified profile, records per-building outcomes and derives stock summaries. The application supports reproducibility and inspection; it does not turn uncertain input data into observation.

Do not describe a BSEW result as:

- measured consumption or a utility-bill observation;
- calibration of an individual building unless a separate calibration protocol supplies and evaluates measurements;
- an energy certificate or regulatory compliance decision;
- embodied carbon or whole-life carbon;
- exact attribution of HVAC or domestic hot-water energy to residential versus commercial space;
- a causal estimate of an intervention unless the study design establishes that interpretation.

Annual runs report an annual simulation period. Microclimate event runs report their recorded event date range and duration; they must not be relabelled `/year`. Operational carbon uses recorded final-energy factors and is not a measured emission inventory.

2.1 What validation means here

The verified model profile is checked against the project's reference-model evidence and fixed acceptance gates. This supports consistency of the modelling method. It does not mean that every stock building was validated against its own measurements. State this distinction in abstracts, captions and results sections.

The documented software baseline is Python 3.13, OpenStudio 3.11.0 with EnergyPlus 25.2.0, and Node.js 20.19 or newer [1]–[4]. Recording those versions supports method reproducibility; using the specified versions is not itself evidence of physical accuracy.

3. Research design before opening the application

Write a short run protocol before using Files:

1. research question and intended comparison;
2. city, district or explicit building scope;
3. annual or event period;
4. input versions and authorised sources;
5. verified profile fingerprint;
6. exclusion and retry policy;
7. retention setting and storage budget;
8. success, QA and diagnostic acceptance rules;
9. planned exports and citation package.

Choose a run name that is unique, stable and meaningful without embedding personal information. Never reuse a completed run name for a different scientific intent. Resume is for the same identity; a changed input, scope or profile requires a new run.

3.1 Worked-example identity

The final example is `ALL_VALENC-A_REAL`, a full Valencia annual stock run. Its final scope, completed counts, period, profile fingerprint, aggregate values and capture date will be read from settled run evidence after its process ends. Interim API summaries are running totals and are unsuitable for publication.

3.2 Reproducibility notebook

Maintain a project log beside the software evidence. Record decisions that the application cannot infer: why the scope was chosen, why exclusions are acceptable, whether a run supersedes another, who reviewed diagnostics, and which export entered the analysis dataset.

4. Files: establish authoritative inputs

The Files page manages four input classes: building GIS, the Tipo15 companion table, the EPW/DDY climate pair, and the OpenStudio template. Imported content is copied into managed storage and identified by content fingerprints. The visible filename aids recognition; the fingerprint establishes identity.

For each input card:

- confirm that the expected item is active;
- read the validation result rather than relying on file extension;
- record the snapshot fingerprint in the study log;
- retain the original supplier and licensing record outside the application;
- investigate warnings before preflight.

Replacing an active input affects future runs only. Existing runs retain their resolved identities. This separation is essential: a publication must cite the inputs used by the reported run, not simply the files active today.

4.1 Input interpretation limits

Cadastral records are administrative evidence, not direct geometric surveys or live occupancy sensors. Padrón population is administrative population evidence, not real-time presence. The template provides constructions, schedules and space-type assumptions; it is not a measurement of each building. Weather files represent the selected climate evidence and must be paired correctly with design-day data.

Final figure 1. Annotated Files page showing the four active input classes, validation state and snapshot identity. Local paths will be removed; the caption will distinguish filename from fingerprint.

5. Preflight: test the proposed scope without running

Preflight is the mandatory gate between input readiness and execution. It evaluates the requested scope, counts buildings in scope, identifies runnable and excluded records, estimates duration and storage, and checks the profile and input identities required by the runner.

Read every preflight field as evidence about the proposed run:

Table 1. Preflight: test the proposed scope without running

Field	Meaning	Interpretation limit
In scope	Records selected by the scope definition	Not all are necessarily runnable.
Runnable	Records that pass current preparation gates	Not a guarantee of simulation or QA success.
Excluded	Records rejected before simulation with reasons	Exclusion is not zero energy.
Estimated time	Projection from comparable recorded runs	Hardware contention and tail cases can change elapsed time.
Storage estimate	Expected protected output under retention settings	Free-space checks and later exports still matter.

Do not start when the resolved scope differs from the protocol. Save or transcribe the preflight counts and profile fingerprint. For full-city work, use the explicit acknowledgement only after reviewing time, storage and interruption recovery.

6. Run configuration and start decision

The Run page turns a reviewed proposal into a named execution. Set scope, workers, retention and run name deliberately. Higher worker counts can reduce elapsed time but increase contention, memory pressure and simultaneous I/O. Use the project-tested setting rather than maximising the number.

Retention controls which per-building artefacts remain available. Full retention supports the strongest audit trail but requires substantial storage. A lighter retention setting may be appropriate for exploratory work only when the research protocol accepts its evidence limits.

Before selecting **Start run**, confirm:

- the run name is unique;
- preflight matches the intended scope;
- the verified profile fingerprint is recorded;
- the period is correct;
- storage and power management are adequate;
- no planned update, build or service restart will occur during execution;
- the recovery procedure is understood.

Starting a run freezes resolved identities. A later interface refresh does not redefine the running work. Do not edit motor-imported source code during a protected stock run because profile-integrity checks may detect drift and invalidate work.

Final figure 2. Annotated Run configuration for `ALL_VALENC-A_REAL`, captured without exposing local paths.

7. Monitor a running stock simulation

While a run is active, the Outputs summary may show completed-building totals. These are running totals, not a forecast of the final aggregate. The correct visual treatment must explicitly say that the run has not finished and that figures can rise as additional buildings complete.

Monitor:

- `running` state and process identity;
- completed, successful, failed and excluded counts;
- recent log activity;
- disk availability;
- machine power and attached storage;
- unexpected profile or input mismatch messages.

Do not infer no bias from a high completion percentage. Missing buildings may differ systematically in size, typology or geometry. Coverage bias can be measured only when a common denominator is available for both represented and missing buildings; otherwise the correct statement is that truncation is unquantified.

7.1 Safe operating discipline

Do not rebuild frontend assets, restart the service, update code, move the run directory or mutate ledger files during a protected run. Read-only API and filesystem checks are acceptable. If the application is locally served, remember that changing built assets can publish immediately even without restarting the backend.

8. Stop safely and resume the same run

Use **Stop safely** when an active run must pause. The runner finishes or records work according to its safe-stop contract and preserves the durable ledger. Do not use force termination as the normal pause mechanism.

Resume requires the same run name and scientific identity. BSEW compares the requested inputs, profile and scope with the recorded run configuration. A mismatch refusal is evidence protection, not an inconvenience to bypass.

A run is complete only when all relevant conditions are settled:

1. the run reports `running: false`;
2. the recorded process no longer exists;
3. the final aggregate exists and is readable;
4. the ledger has settled to the expected scope accounting;
5. final exports are generated from that settled evidence.

Process absence without a final aggregate means interruption or failure, not completion. In that state, inspect the log and resume with the same identity when appropriate. Never rename or move the original run root merely to make it look finished.

8.1 Failure and retry

Retry failed records only under a documented policy. Pre-simulation exclusions and simulation failures are different classes and should remain distinguishable. A changed model gate usually implies a new profile and new run, not silent replacement of historical records.

9. Outputs: select one run and establish context

Outputs is an evidence reader. Select a run first; every aggregate, ledger row and export action must refer only to that selected run. The run selector should show status, and an active run should never be deletable.

For a finished run, begin with:

- run identity and period;
- final status and coverage;
- result and QA counts;
- verified profile fingerprint;
- principal area denominators;
- operational-emissions basis;
- warnings about exclusions, partial coverage or classified diagnostics.

The toolbar provides user-facing exports such as Building CSV, GIS layer, heat map and a signed package. These are different representations of the same run evidence, not interchangeable scientific objects. A PNG communicates spatial patterns; it cannot replace the row-level CSV. A CSV supports analysis; it does not preserve every model and diagnostic file. A signed package supports integrity; it does not make an interpretation correct.

Final figure 3. Completed `ALL_VALENC-A_REAL` Outputs header with the selected run, orderly action toolbar and final status.

10. Metric reading contract

Every reported metric should be read through six fields: visible name, authoritative source, unit, period, denominator and interpretation limit. BSEW does not recompute physical results in the readable report; it may format units and labels and check consistency among preserved evidence blocks.

Table 2. Metric reading contract

Visible metric	Authoritative source	Unit and period	Denominator	Interpretation limit
Total site energy	Building `results` records aggregated for successful rows	kWh/year or GWh/year for annual runs	None for the total	Modelled final site energy, not measured utility use.
Whole-site energy over geometric residential area	Recorded aggregate/result fields	$kWh/m^2 \cdot year$	Geometric residential floor area	Includes all site energy in the numerator; use for the declared comparison convention only.
Whole-site energy over cadastral area	Recorded aggregate/result fields	$kWh/m^2 \cdot year$	Cadastral Tipo15 residential area	Sensitive to cadastral/modelled area allocation.
Energy over conditioned area	Recorded results	$kWh/m^2 \cdot year$	Total conditioned model area	Different from the residential-only denominators.
Operational carbon	Recorded carbon block	tCO_2/year or event-period tCO_2	None for the total	Operational factors only; not embodied or measured emissions.

Missing optional evidence is shown as an em dash. A measured zero remains `0`. Invalid types, booleans as numbers, NaN and infinity must not be printed as valid metrics.

11. Area denominators and why they differ

One building-stock result can legitimately have several intensities because the numerator and denominator answer different questions. Never shorten all of them to “EUI” without naming the area basis.

Geometric residential area is the modelled residential-storey area used by the comparison convention. **Cadastral Tipo15 area** is the administrative residential-area basis when available. **Conditioned area** is the area actually conditioned in the model, including applicable non-residential space.

The whole-site numerator can include commercial ground-storey energy while a residential-only denominator excludes that floor. This is a deliberate comparison convention, not the physical intensity of all conditioned space. Report both the convention and its limit.

Integer storey modelling can exceed a fractional cadastral area because an actual model needs whole storeys. The recorded top-storey fraction scales dwelling loads on the excess portion. The aggregate's floor-area allocation section describes the recorded rule and hypothetical alternatives; it must not rewrite a historical run using today's model rule.

When comparing cities, clusters or methods, require the same period, numerator and denominator. Similar numeric values under different bases are not equivalent observations.

12. Energy end uses and residential attribution

EnergyPlus Annual Building Utility Performance Summary values come from output meters. Subcategory splits depend on the model's user-defined end-use subcategories [5]. Use the preserved EnergyPlus result tables to inspect end uses and the readable report to understand how BSEW labels them.

Space heating, cooling and domestic hot water are model outputs under the recorded assumptions. Electricity and gas contribute to total site energy. Rounding in exported EnergyPlus tables can create small differences when individually rounded cells are summed; compare against authoritative preserved totals at their recorded precision.

`residential_site_kwh` is not an exact residential subtotal. Commercial lighting and equipment can be separated by subcategory, but HVAC and domestic hot water are not fully separable by space type in the preserved meter contract. The correct label is **residential-attributed upper bound**. Do not call it measured residential consumption or a physical residential EUI.

When there is no commercial space, the report should not claim commercial consumption merely because a metering configuration exists. When the ground storey is residential, use neutral language such as “ground-storey openings”; reserve “shopfront” for a commercial ground regime.

13. Operational carbon

BSEW records operational emissions derived from final-energy results using project factors. The current verified contract uses 0.331 $kgCO_2$ per kWh of electricity and 0.252 $kgCO_2$ per kWh of final natural gas, documented against the cited IDAE technical source [6].

Interpretation limits:

- the result is operational, not embodied or whole-life carbon;
- it is calculated from simulated final energy, not measured emissions;
- factors represent the adopted evidence basis and may not match a different year, supplier or marginal-emissions question;
- annual runs use `tCO₂/year`;
- event runs use **event-period operational emissions** and never `/year`;
- an unrecorded period makes otherwise successful evidence incomplete for citation.

Report factors beside the result or in the methods section. Do not convert them into a claim of certification or regulatory compliance. If a future study uses different factors, it should preserve the original energy result and clearly version the derived carbon method.

13.1 Domestic hot water assumption

The cited CTE reference demand of 28 litres per person per day is specified at 60 degrees Celsius [7]. A model implementation at 50 degrees Celsius is not automatically equivalent. State both the reference and implementation temperatures and avoid claiming direct CTE equivalence.

14. Building CSV

The user-facing **Building CSV** is intended for analysis in spreadsheet, statistical and GIS workflows. The final curated export will use a stable academic schema and be accompanied by a data dictionary. It will remain distinct from the backward-compatible raw ledger CSV.

For every column, the dictionary should state:

- visible column name;
- authoritative ledger source field;
- data type;
- unit;
- period;
- denominator, when applicable;
- missing-value rule;
- interpretation limit.

The export must preserve the status of every in-scope reference. Excluded and failed records are not silently dropped or converted to zero. Numeric columns must remain numeric; measured zero and missing evidence must remain distinct. Boolean and categorical values should use a documented representation.

Before analysis, verify run identity, row count, unique building reference, status counts, period and profile fingerprint against the selected run. Keep the downloaded file hash in the analysis log. If the curated export is regenerated from a settled ledger, no EnergyPlus rerun is required; the transformation must be deterministic and tested against the ledger's latest record per reference.

Final figure 4. Building CSV download and a short annotated excerpt with units and dictionary linkage.

15. GIS layer

The GeoPackage export joins building geometry to recorded result fields and may include aggregate layers such as cluster or district summaries and style metadata. Use it when spatial analysis requires vector geometry, feature-level attributes or reproducible GIS styling [9].

Check before use:

- coordinate reference system;
- geometry validity and feature count;
- one-to-one relationship between building reference and latest run record;
- status representation for successful, excluded and failed records;
- units and denominators of mapped attributes;
- field-name mapping to the Building CSV/data dictionary;
- whether any geometry lies outside a chosen display frame.

A GIS layer is not a basemap and should not require an external network service for core evidence. When overlaying third-party boundaries or imagery, cite their source, date and licence separately.

Aggregate district or cluster intensities should use the declared area-weighted contract. Do not average per-building intensities arithmetically unless that is the research question. If a district layer is empty because the required district field was not available, report it as unavailable rather than a district total of zero.

16. Heat map

The heat map is a publication-oriented spatial summary derived from the settled result geometry. It must identify the run, period, metric, denominator, colour-scale treatment, coverage and simulation boundary.

For annual Valencia evidence, panels may show whole-site energy over geometric residential area, space heating and cooling in kWh/m²·year. Event runs must state the event duration and must not be annualised by label. A metric constant across the entire stock should be omitted with an explicit note rather than rendered with a misleading colour range.

Percentile clipping, such as P2-P98, preserves visible contrast but does not change underlying values. The final map should disclose numeric limits and counts below/above the display range. Successful, excluded, failed and metric-missing records require a separate status legend.

Disconnected or distant valid geography must not disappear from the figure. The planned final Valencia map will use a main view and evidence-preserving inset logic so every valid building is visible at least once. It will state the CRS, profile fingerprint and “simulated, not measured consumption” boundary.

Final figure 5. Corrected `ALL_VALENC-A_REAL` heat map after the run settles; no interim map will be cited.

17. Readable building report

Select a building row to open the readable building report. The report is the primary human-readable bridge between stock totals and preserved technical files. It uses the authoritative `results`, `carbon` and `qa` blocks; duplicate summary fields are consistency checks, not alternate sources.

The report should present:

1. identity, period and publication status;
2. interpretation summary;
3. scope and limitations;
4. energy and emissions results;
5. geometry and occupancy;
6. QA and diagnostics;
7. model preparation and assumptions;
8. provenance, references and technical evidence.

If protected evidence blocks conflict, the report must show **EVIDENCE CONFLICT — DO NOT CITE** rather than silently selecting a value. A route/ledger/building identity mismatch should prevent report production. A successful-looking record without essential period, result or QA evidence is **INCOMPLETE EVIDENCE — DO NOT CITE**.

The readable report is academic English with an A4 print layout. Its screen presentation uses semantic headings, labelled links, keyboard focus and a WCAG 2.2 AA accessibility target [10]. Absolute local filesystem paths are excluded. Identifiers and fingerprints remain visible because they establish reproducibility.

Final figure 6. Building report for a real `ALL_VALENC-A_REAL` reference, including interpretation, metric basis and QA status.

18. EnergyPlus result tables and technical files

EnergyPlus result tables are directly visible in Readable Evidence because they contain the original annual simulation tables, including areas, end uses, zone results and comfort evidence. Read them together with the report's definitions and limitations.

The original technical files remain available for specialist audit:

- `eplustbl.htm`: EnergyPlus HTML result tables;
- `eplusout.err`: warnings, severe and fatal diagnostic log;
- `model_python.osm`: generated OpenStudio model;
- `deep_layers.json`: machine-oriented record of preparation stages and checks;
- geometry scene/model evidence where available.

These files are not hidden evidence; they are separated because their native formats are designed for engines and specialists. The readable report must not replace or alter them. If a publication depends on a precise end-use value or diagnostic classification, cite the readable explanation and retain the original technical file in the audit package.

Do not describe a technical object name such as `1th floor` as academic prose. If a raw identifier must be shown, label it explicitly as a raw model identifier and format it as code.

19. QA decision matrix

BSEW uses explicit publication statuses:

Table 3. QA decision matrix

Status	Required evidence	Permitted interpretation
FAILED / DO NOT USE	Failed run result, QA false, Fatal, or unexplained Severe	Do not cite as an accepted building result.
ACCEPTED WITH CLASSIFIED DIAGNOSTICS	Successful result; recorded checks pass; no Fatal/unexplained Severe; project-classified benign Severe exists	May be used only with the project-specific diagnostic classification disclosed.
PASS	Successful result; QA passed; no Severe or Fatal	Accepted under the recorded QA contract.
NOT ASSESSED	No sufficient QA decision was recorded	Do not infer pass.
INCOMPLETE EVIDENCE — DO NOT CITE	Essential result, period, QA or diagnostic evidence is missing	Do not cite until evidence is complete.

Status is communicated with words and symbols, never colour alone. PASS does not mean measured calibration or universal model validity; it means that the recorded checks and diagnostic gates passed.

19.1 Classified Severe messages

EnergyPlus generally treats Severe messages as conditions requiring correction [8]. A project may classify a narrowly measured pattern as benign under an explicit acceptance rule. **ACCEPTED WITH CLASSIFIED DIAGNOSTICS** makes that exception visible. Review the actual message markers and the classification basis; do not shorten the status to PASS.

20. QA checks and tolerances

The readable QA table uses human-readable check names beside machine keys. Its columns distinguish expected/model value, observed value or accepted interval, acceptance criterion and result.

Key tolerances are reported in reader-facing units:

- a stored ratio tolerance of `0.005` is shown as `±0.5%`;
- `0.02` is shown as `±2%`;
- annual unmet hours are accepted only when the recorded value is `≤500 h`.

Some checks compare the built model with EnergyPlus-reported evidence. A plausibility-band check is different: it evaluates the method against a declared research range and must not be described as an EnergyPlus cross-check.

Review failed checks at building level and in aggregate. A zero QA-failure count does not resolve exclusion bias, input uncertainty or model-form limitations. Conversely, a failed or excluded building is not evidence of zero demand.

When exporting a QA table, preserve the original key for reproducibility but lead with the readable name and criterion. Do not leave naked machine keys as the only academic explanation.

21. Geometry, floors and occupancy

The model simplifies cadastral geometry into a tractable building representation. The report distinguishes total residential levels, effective residential storeys and any storey-rule adjustment. These terms are not interchangeable.

Ground-storey interpretation is conditional on the recorded final use:

- a residential ground storey is a dwelling space, thermostat-controlled and included in the residential area basis;
- a commercial ground storey follows the verified tertiary-use regime and is excluded from the residential-only denominator;
- mixed or missing evidence must be described without forcing either narrative.

Occupancy reporting separates the administrative Padrón value, the person count applied to the model, any density cap, and cluster-median imputation. Padrón is not real-time occupancy. A zero recorded population, an imputed value and a capped value are distinct evidence states.

Footprint simplification fidelity and storey snapping are recorded quality fields. A high-fidelity score does not establish survey accuracy; it describes the transformation relative to the supplied geometry. Buildings without a usable exterior wall may legitimately have zero glazed subsurfaces, which should be stated explicitly rather than treated as missing data.

22. Provenance and reproducibility

A citable result needs more than a run name. Preserve:

- run identity and final status;
- input dataset filenames and content fingerprints;
- verified profile fingerprint;
- software commit;
- climate/template/policy/stock-source/runner-schema identities;
- run configuration and process record;
- durable ledger and final aggregate;
- EnergyPlus/OpenStudio versions;
- generated CSV/GIS/heat map hashes;
- selected building reports and original technical evidence;
- review date and reviewer decision.

The report should use logical dataset names, filenames and hashes, not absolute local paths. Unpublished research-group evidence must be labelled as such with filename, verification date and fingerprint; do not invent bibliographic metadata.

22.1 Consistency checks

When the same metric is preserved in both authoritative and summary blocks, compare them. A contradiction is a reportable evidence conflict. When rebuilding an aggregate from a ledger, verify that all latest records share the intended profile identity; a mechanically combined multi-profile ledger must not appear as one homogeneous run without an explicit warning.

Reproducibility means another authorised researcher can identify and inspect the same evidence and method. It does not guarantee identical wall-clock time or remove uncertainty from the original data.

23. Export, integrity and retention

Use the smallest export that preserves the evidence needed for the research claim:

- Building CSV plus dictionary for row-level statistical analysis;
- GeoPackage and style for spatial analysis;
- heat map for communication, never as the sole numeric evidence;
- signed building package for a focused audit;
- full signed ZIP only when storage, duration and purpose justify it.

A signed manifest verifies that package contents match the recorded manifest and signing key. It does not certify scientific validity. Verify signatures and hashes after transfer and retain the verification log.

Do not delete a real run during acceptance testing. User-controlled deletion requires the exact run name and must be refused authoritatively while a process is active. Before deleting a finished run, verify independent backup, export integrity, citation dependencies and retention obligations.

The planned final result folder will be a verified copy outside the original run root. The original `out/stock/ALL_VALENC-A_REAL/` remains in place so run identity, routes and signed export sources are not broken. Copy acceptance requires source-to-target SHA-256 equality.

24. Reporting checklist for academic use

24.1 Methods section

Report BSEW name and commit; Python/OpenStudio/EnergyPlus versions; run name; input and profile fingerprints; scope; period; worker/retention configuration where relevant; principal modelling assumptions; exclusion policy; QA criteria; energy denominators; carbon factors; and the logical evidence-package identifier or repository location. Do not publish a local absolute path.

24.2 Results section

Report final successful, failed and excluded counts; coverage with an honest bias statement; total energy and named intensity bases; operational carbon with period; cluster/district method; diagnostics status; and uncertainty or limitation that materially affects interpretation.

24.3 Figures and tables

Every figure caption should include run, period, metric, unit, denominator, coverage and simulated-not-measured boundary. Every table should identify the source export and hash. Explain percentile clipping and missing-status encoding on maps.

24.4 Minimum review before citation

- final process gone and aggregate settled;
- run/profile/input identities consistent;
- no evidence conflict or incomplete-citation status;
- QA matrix reviewed, including classified diagnostics;
- CSV/GIS/map counts reconciled to the ledger;
- hashes and manifest recorded;
- claims use simulation language and correct denominators;
- local paths, personal information and unauthorised source files absent.

25. Limitations and responsible interpretation

The final Valencia publication must retain these boundaries where relevant:

- cadastral geometry and area are administrative inputs with measurement and allocation limits;
- a parcel is not always a single physical building;
- integer-storey modelling and simplified zoning affect floor-area representation;
- large one-zone buildings may not represent core/perimeter diversity;
- courtyard and difficult geometry can be excluded by current preparation rules;
- occupancy uses administrative evidence, caps and, where recorded, imputation;
- HVAC and DHW cannot be cleanly split between residential and commercial space in the preserved meter contract;
- operational factors do not include embodied carbon;
- one verified profile does not constitute per-building calibration;
- LHS evidence for a pilot building is not a city-total uncertainty band;
- failed and excluded records can create unquantified truncation when a common area basis is absent.

Limitations are not an appendix to be ignored. Place the relevant limitation beside the metric, figure or comparison it constrains. A concise visible warning is more scientifically useful than a long disclaimer disconnected from the result.

26. References and document control

[1] Python Software Foundation, "Python 3.13 documentation," https://docs.python.org/3.13/

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[3] EnergyPlus, "EnergyPlus 25.2 documentation," https://energyplus.readthedocs.io/en/v25.2.0/

[4] OpenJS Foundation, "Node.js downloads," https://nodejs.org/en/download

[5] U.S. Department of Energy, *EnergyPlus Input Output Reference*, version 25.2; HTML rendering of the release documentation by Big Ladder Software, https://bigladdersoftware.com/epx/docs/25-2/input-output-reference/index.html

[6] Instituto para la Diversificación y Ahorro de la Energía, *La bomba de calor en la rehabilitación energética de edificios*, 2023, https://www.idae.es/sites/default/files/documentos/publicaciones_idae/Guias_IDAE_La_Bomba_de_calor_2023_V11.pdf

[7] Gobierno de España, *Código Técnico de la Edificación, DB-HE*, 2017 edition, https://www.codigotecnico.org/pdf/Documentos/HE/DBHE_201 706.pdf

[8] U.S. Department of Energy, *EnergyPlus Essentials*, version 25.2, https://energyplus.readthedocs.io/en/v25.2.0/essentials/essentials.html

[9] Open Geospatial Consortium, "GeoPackage Encoding Standard," https://www.ogc.org/standard/geopackage/

[10] W3C, "Web Content Accessibility Guidelines (WCAG) 2.2," https://www.w3.org/TR/WCAG22/

Document control

This provisional edition is generated from one canonical source shared by HTML and DOCX. PDF is produced from the visually verified DOCX. The final edition will remove the provisional status, add the settled `ALL_VALENC-A_REAL` evidence and annotated real screenshots, update the release commit and publication manifest, then pass cross-format heading, figure, reference and citation equality checks.

The final manifest will record canonical-source hashes, software commit, publication date and SHA-256 for each DOCX, PDF and HTML deliverable. The living HTML will link to this Installation Guide and the application's Help menu; date-versioned DOCX/PDF files remain immutable publication artefacts.